

Milestone 3.1 - First Sprint: Task Planning and Estimation

ECE444 Software Engineering · Project 1: Web Application Development

Team Nexus (Group 20) · What's in My Food?

Members: Suyeon Park, Frank Liu, Eyad Ahmed, Alex An

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Location:

<https://github.com/suyeon240park/ECE444-Group20/blob/main/docs/M3-sprint-1-report.pdf>

1. Work backlog (2 pts)

Tool: GitHub Projects for the board, GitHub Issues for stories and tasks, GitHub Pull Requests for code review.

- Board: <https://github.com/users/suyeon240park/projects/2>
- Issues: <https://github.com/suyeon240park/ECE444-Group20/issues>
- Requirements: <https://github.com/suyeon240park/ECE444-Group20/blob/main/docs/requirements.md>

How it is organised:

- The 7 user stories selected in Milestone 2 are issues labelled **candidate-story**: #10 (F1), #2 (F2), #3 (F3), #9 (F4), #4 (Q1), #5 (Q2), #11 (Q3).
- Each story has a **Priority** (High / Medium / Low) and a **Sprint** (Sprint 1 / Sprint 2 / Backlog). The highest-priority stories, #10 and #4, are in development in sprint 1.
- Each story is split into task issues with an owner, which move Todo → In Progress → Done.

Sorted user stories (filtered to the 7 stories, grouped by sprint, sorted by priority):

ECE444 Group 20 — Backlog

View 1

Backlog

+ New view

Q label:candidate-story

Title

...

Assignees

...

Sprint

...

Priority

...

Labels

...

▼

🟢 Sprint 1

2

Current sprint

...

1

🟢 Identify a packaged food from an uploaded photo and retrieve its food information #10

▼

2

🟢 Nutrition values extracted from a label photo must match the printed label #4

▼

+

Add item

▼

🟡 Backlog

5

Selected, not yet scheduled

...

3

🟢 Display calories and nutrients for the entire package #2

▼

4

🟢 Explain each nutrient on a product in plain language with a credible source #3

▼

5

🟢 Identify uploaded food products within an acceptable response time #5

▼

6

🟢 Present food analysis that non-experts can understand and navigate without help #11

▼

7

🟢 Quick ingredient summary for busy shoppers #9

▼

+

Add item

Sprint 1

High

candidate-story

functional

Sprint 1

High

candidate-story

quality

Backlog

High

candidate-story

functional

Backlog

High

candidate-story

functional

Backlog

High

candidate-story

quality

Backlog

High

candidate-story

quality

Backlog

Medium

candidate-story

functional

2. Sprint 1 summary and Sprint 2 plan (5 pts)

2.1 Sprints

Sprint	Dates	Goal
1	October 3 – October 9	A deployed skeleton: submit a product photo, detect the barcode, get Open Food Facts data, show complete / partial / not-found results.
2	October 10 – October 23	The user-facing product: whole-package nutrition, nutrient explanations, ingredient analysis, one results page.
3	October 24 – November 20	Quality and release: performance, usability, end-to-end tests, bug fixes, final deployment.

Each sprint ends on the course's progress-update deadline. Final delivery and demo: December 1.

2.2 Sprint 1 summary

Decisions

- **Tech stack:** Expo / React Native (TypeScript) for web, iOS and Android; Flask backend; PostgreSQL; Open Food Facts for product data; GitHub Actions for CI.
- **Photo path kept:** we replaced OCR with a vision-model API. The prototype (#16) read all 166 values on 10 label photos correctly with `gemini-3.5-flash` (free, about 7 s per photo).
- **API contract (#20):** one response format for `GET /api/products/{barcode}`. Missing data is `null`, never 0.
- **Staging:** Render free tier, deployed automatically after tests pass on `main` (#22).

Sprint 1 tasks

Issue	Task	Owner	Est. hours	Status
#19	Project skeleton, CI, dev environment	Frank	6–8	Done (PR #24)
#23	README, CONTRIBUTING, Code of Conduct	Frank	3–5	Done (PR #21)
#16	Vision-model label-reading prototype	Frank	5	Done (PR #27)
#22	Staging deployment	Frank → Eyad	5–8	In progress (code done; Render setup in sprint 2)
#20	Product data model and API contract	Suyeon	4–6	Done (PR #25)
#14	Open Food Facts product lookup	Suyeon	6–8	Done (PR #26)

Issue	Task	Owner	Est. hours	Status
#13	Photo upload and barcode detection	Eyad	6–10	Done (PR#30, PR #32)
#29	Final review of the Milestone 3 document	Eyad	1	In progress
#28	Open Food Facts coverage check	Alex	5–8	Done (results in #28)
#15	Completeness check and photo fallback routing	Alex	5–8	Done (PR #31)

What did not go as planned

- **Too much waiting:** most tasks needed the project skeleton, which only merged on October 8. Next time the skeleton comes first and other work starts with mock data.
- **Estimates:** coding tasks took less time than estimated, but code review took more (for example 4 h against 1–2 h planned). Sprint 2 budgets more time for reviews.
- **Planned issue numbers clashed** with pull-request numbers, so we now create issues before referring to them.
- **Midterms** reduced everyone's available hours.

Open questions for sprint 2

- **Answered by #28:** Open Food Facts found 29 of 30 Canadian products, but only 9 had everything the app needs, and package size was missing or wrong for most. The app will ask the user for package size, and incomplete products go to the photo path.
- Do we store uploaded photos? On the free tier, Google may use the photos to improve its models, so the app should say so.
- Photo values need two additions to the API contract: a certainty flag on serving size, and a "confirmed by user" state.

2.3 Sprint 2 plan

Goal: whole-package nutrition, nutrient explanations, ingredient analysis, and one results page.

Issue	Task	Story	Est. hours	Depends on	Owner	Reviewer	PR by
#33	Ingredient knowledge base: schema, parsing, matching	#9	12	—	Suyeon	Frank	Oct 13
#34	Ingredient entries: common ingredients (sugar, salt, oils, flours, starches, emulsifiers)	#9	6	#33	Suyeon	Alex	Oct 16
#35	Ingredient entries: additives that may be flagged (colours, preservatives, sweeteners)	#9	6	#33	Suyeon	Alex	Oct 16
#36	Whole-package nutrition calculation and display	#2	10	—	Alex	Suyeon	Oct 15

Issue	Task	Story	Est. hours	Depends on	Owner	Reviewer	PR by
#37	%DV interpretation and nutrient explanations	#3	10	—	Alex	Suyeon	Oct 15
#38	Reviewed nutrient explanation content	#3	8	—	Alex	Frank	Oct 15
#39	Screen to confirm photo values before use	#4, #10	6	#42	Alex	Eyad	Oct 17
#40	Ingredient summary and detail screens	#9	8	#33	Eyad	Suyeon	Oct 17
#41	Results page	#2, #3, #9	12	#43	Eyad	Alex	Oct 18
#42	Photo endpoint using the #16 prototype	#10, #4	6	#15, #16	Frank	Eyad	Oct 14
#43	Pipeline that combines all analyses	#10, #2, #3, #9	12	#33, #36, #37	Frank	Suyeon	Oct 17

Issue	Task	Story	Est. hours	Depends on	Owner	Reviewer	PR by
#17	50-photo label benchmark, checked by Alex	#4	6	#16	Frank	Eyad	Oct 18
#44	End-to-end tests for nutrition and ingredients	#2, #3, #9	6	#41, #43	Frank	Eyad	Oct 19
#45	Loading, partial-data, empty and error states (moves to sprint 3 if time runs out)	#10, #2, #3, #9	6	#41	Eyad	Alex	Oct 19

Carried over from sprint 1: #22 staging (Eyad, reviewed by Frank)

Every task has a GitHub issue with a description, "Done when" criteria and its reviewer. Each member reviews four tasks, never their own.

3. Individual task tables

Estimated hours were recorded before starting; actual hours are the time spent. For unfinished tasks, the actual is the time spent so far.

3.1 Liu's tasks

Liu's tasks for the 1st sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#19 Project skeleton, CI and dev environment (incl. review fixes)	7	1.5
#23 README, CONTRIBUTING and Code of Conduct *	4	1
#16 Vision-model label-reading prototype (time-boxed)	5	2.5
#22 Staging deployment (code and setup guide; handed to Eyad)	6	0.5
Set up and sort the backlog on the board	2	0.5
Draft this sprint report	2	2
Review team-member pull requests timely and thoroughly (PR #25, PR #26)	2	0.5
Participate actively in common team duties (meetings, planning, Discord)	2	1.5

* Written October 4, before sprint planning; the estimate is from the October 5 planning table.

Liu's tasks for the 2nd sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#42 Photo endpoint using the #16 prototype	6	
#43 Pipeline that combines all analyses	12	
#17 50-photo label benchmark	6	

Task	Estimated Time (hour)	Actual Time (hour)
#44 End-to-end tests for nutrition and ingredients	6	
Review team-member pull requests timely and thoroughly	3	
Participate actively in common team duties	2	

3.2 Park's tasks

Park's tasks for the 1st sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#20 Product data model and API contract	4–6	2
#14 Open Food Facts product lookup	6–8	4
Sprint planning tables and task breakdown	0.5–1	1
Review team-member pull requests timely and thoroughly (PR #21, #24, #30)	1–2	4
Participate actively in common team duties	1–2	2

Park's tasks for the 2nd sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#33 Ingredient knowledge base: schema, parsing, matching	12	

Task	Estimated Time (hour)	Actual Time (hour)
#34 Ingredient entries: common ingredients	6	
#35 Ingredient entries: additives that may be flagged	6	
Review team-member pull requests timely and thoroughly	3	
Participate actively in common team duties	2	

3.3 Ahmed's tasks

Ahmed's tasks for the 1st sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#13 Photo upload and barcode detection	6–10	3
Sprint report and backlog screenshot	1–2	0
Final review of the Milestone 3 document (#29)	1	0
Review team-member pull requests timely and thoroughly	1–2	0
Participate actively in common team duties	1–2	2
#32 Barcode screen: upload a photo or scan live	6	2

Ahmed's tasks for the 2nd sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#40 Ingredient summary and detail screens	8	
#41 Results page	12	
#45 Loading, partial-data, empty and error states	6	
#22 Finish staging deployment (Render setup, deploy secrets, live URLs)	3	
Review team-member pull requests timely and thoroughly	3	
Participate actively in common team duties	2	

3.4 An's tasks

An's tasks for the 1st sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#28 Open Food Facts coverage check	5–8	5
#15 Completeness check and photo fallback routing	5–8	5
Review team-member pull requests timely and thoroughly (PR #30)	2	0.5
Participate actively in common team duties	2	1

An's tasks for the 2nd sprint include:

Task	Estimated Time (hour)	Actual Time (hour)
#36 Whole-package nutrition calculation and display	10	
#37 %DV interpretation and nutrient explanations	10	
#38 Reviewed nutrient explanation content	8	
#39 Screen to confirm photo values before use	6	
Check the ground truth of the 50-photo benchmark (#17)	1	
Review team-member pull requests timely and thoroughly	3	
Participate actively in common team duties	2	